

PerspectEvolver: Evolving Perspectives through Multi-Agent Deliberation for Research Ideation

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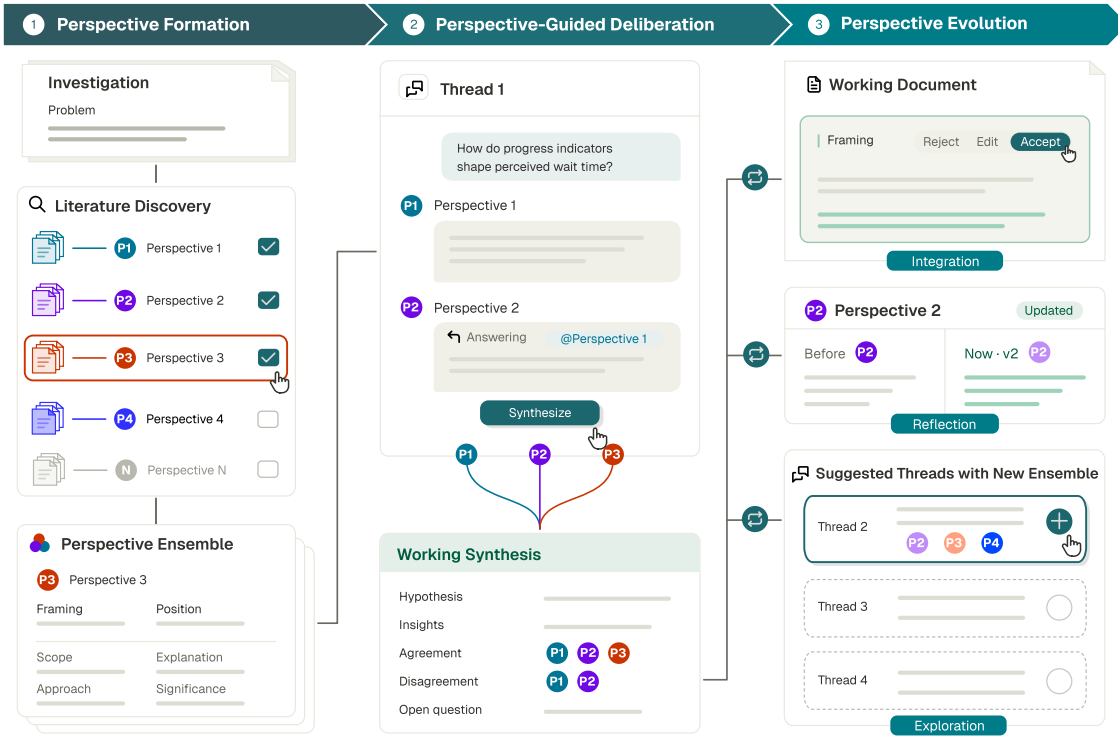


Fig. 1. *PerspectEvolver* interface. The system supports sensemaking around the configuration and evolution of perspectives as deliberation develops through three main components: ① **Perspective Formation**, where each perspective is built from a cluster of papers and represented as six inspectable elements: Framing, Position, Scope, Explanation, Approach, and Significance; ② **Perspective-Guided Deliberation**, where the perspectives answer one another in a topic thread, which Synthesize condenses into a Working Synthesis recording the hypothesis, insights, agreements, disagreements, and open questions the thread produced; and ③ **Perspective Evolution**, where that synthesis is carried in parallel into the Working Document (integration), elements of perspectives (reflection), new threads (exploration), for which the user assembles a new perspective ensemble from evolved and formed perspectives in Perspective Formation.

Research ideation involves exploring alternative ways of framing, explaining, and investigating a problem. Collaboration expands this exploration as different perspectives come together, allowing new ideas and directions to emerge. Current multi-agent systems

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(MAS) support ideation through diverse agent perspectives, but offer limited support for understanding how perspectives are formed, how they contribute to research ideas, and how they evolve. We present *PerspectEvolver*, an interactive MAS with three phases: 1) Perspective Formation constructs literature-grounded Perspectives; 2) Perspective-Guided Deliberation develops directions and synthesizes discussions into Working Syntheses; and 3) Perspective Evolution uses these syntheses to revise Perspectives, refine the Investigation Brief, and generate follow-up questions. In a 10-participant within-subjects study, *PerspectEvolver* produced significantly greater improvements in hypothesis novelty than a group-chat baseline. Participants rated *PerspectEvolver* higher for usability and support for divergent and convergent thinking, while reporting significantly lower mental demand and effort. We discuss implications for evolving perspectives throughout research ideation.

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1 Introduction

Developing a research idea requires exploring a vast hypothesis space and refining promising possibilities into hypotheses worth pursuing [39, 43]. Colleagues with different perspectives broaden this exploration by exposing assumptions and introducing possibilities researchers may not consider independently [1, 16, 43]. Those perspectives can also change as new distinctions, explanations, or evidence revise how researchers understand the problem and what they pursue next [25].

LLM-based multi-agent systems increasingly support collaborative ideation by assigning agents different perspectives [29–31, 40]. These perspectives expose researchers to alternative interpretations and can broaden the ideas considered during research ideation [29–31, 40].

However, prior multi-agent systems primarily support configuring diverse agent perspectives, with less support for understanding how those perspectives contribute to and change as the research direction develops [29, 30, 40]. Researchers must reassess the developing research direction, including which perspectives remain relevant, how they have changed, and which questions warrant further investigation. Supporting this ongoing sensemaking around perspective configuration, contribution, and evolution remains a design challenge for multi-agent research tools.

To understand how researchers manage perspectives in AI-supported ideation, we conducted a formative study with six researchers using general-purpose LLMs in their existing workflows. We found three challenges: (1) distinguishing what each perspective represents, (2) understanding how each shapes the developing research direction, and (3) determining how those perspectives should change as that direction develops. Researchers had to reconstruct and update this context manually across interactions.

Building on these findings, we developed *PerspectEvolver* (Figure 1), an interactive multi-agent system that externalizes literature-grounded Perspectives as inspectable and revisable representations. *Perspective Formation* organizes relevant literature into regions and forms Perspectives for researchers to assemble into a Perspective Ensemble. *Perspective-Guided Deliberation* develops research directions through Deliberative Threads and consolidates each discussion into a Working Synthesis. *Perspective Evolution* uses these syntheses to revise Perspectives and the Investigation Brief and generate follow-up questions.

We evaluated *PerspectEvolver* in a within-subjects study with 10 participants from bioengineering, computer science, health sciences, electrical and computer engineering, and mechanical engineering. We compared *PerspectEvolver* against a group-chat multi-agent baseline that retained literature retrieval, multi-agent deliberation, and hypothesis editing, but

omitted structured Perspective representation, attribution of each Perspective’s contribution, Perspective Evolution, and follow-up question generation. We analyzed system logs, survey responses, think-aloud data, interviews, and changes in participants’ submitted hypotheses. Compared with the baseline, *PerspectEvolver* produced significantly greater improvements in hypothesis novelty and received higher usability ratings from participants. Participants also reported lower cognitive load, with significant reductions in mental demand and effort.

Our work contributes to HCI research on multi-agent systems for scientific ideation in three ways:

- **Design and implementation of *PerspectEvolver*.** We introduce an interactive multi-agent system that externalizes literature-grounded Perspectives as inspectable and revisable representations. Researchers form Perspectives from the literature, deliberate with them through Perspective Agents, and revise them as the investigation develops. Deliberative Threads and Working Syntheses preserve how interpretations develop and inform subsequent Perspective revision and inquiry.
- **Empirical evidence from a within-subjects study.** Participants using *PerspectEvolver* developed hypotheses with significantly greater improvements in novelty, while plausibility, significance, and testability also improved more on average but did not differ significantly from the baseline. They also rated *PerspectEvolver* higher in usability and in its support for divergent and convergent thinking, while reporting significantly lower mental demand and effort. Prompt analyses further showed that interactions with *PerspectEvolver* focused more on advancing the hypothesis through GENERATION, EXPLORATION, SYNTHESIS, and SELECTION, whereas baseline interactions focused more on establishing the grounds for those moves.
- **Design implications for perspective-based multi-agent systems.** We identify three design implications: maintaining the context and provenance needed to ground research decisions while preserving researcher judgment over how the inquiry advances; providing an outlet for unresolved disagreement through subsequent exploration; and expanding perspective diversity without proportionally increasing cognitive load.

2 Related work

We review prior work on how perspectives evolve through collaboration and why that evolution requires sensemaking (§2.1), on how multi-agent ideation systems configure and evolve the perspectives they assign to agents (§2.2), and on the divergent and convergent activities that perspective evolution sustains (§2.3).

2.1 Perspective Evolution in Collaborative Hypothesis Generation

Research ideation is an iterative, non-linear process in which ideas emerge, develop, and take new directions as researchers encounter new information and interpretations [39, 43]. As researchers develop hypotheses, they interpret relevant literature through their current understanding of the problem, while new findings can in turn change that understanding [17, 25]. Klein’s data–frame account captures this reciprocal process: a frame guides which information researchers attend to and how they interpret it, while incoming data can prompt them to question and revise that frame [25]. Hypothesis generation therefore depends on both the evidence available and the perspective through which researchers interpret it. Collaboration introduces additional perspectives into this process. Colleagues may approach the same problem with different assumptions, explanations, and criteria, drawing attention to relationships or limitations that are less visible from a single framing [43]. Working through these differences can change both the hypotheses researchers develop and the perspectives through which they understand the problem. As perspectives evolve, they influence what researchers consider next, creating new starting points for further ideation.

Table 1. Comparison with multi-agent ideation systems along how perspectives are configured for the agents, whether each perspective’s contribution to the deliberation is surfaced, and whether perspectives evolve with the deliberation. ● = supported; ◐ = partial; – = not supported.

System	Perspective configuration for multi-agents	Perspective contribution in deliberation	Perspective evolution with deliberation
MultiColleagues [40]	Role-based	◐ (attribution only)	–
CritiqueCrew [6]	Role-based	◐ (attribution only)	–
NarrativeLoom [31]	Role-based	–	–
PersonaFlow [30]	Domain-based	–	–
Perspectra [29]	Domain-based	◐ (attribution only)	◐ (implicit)
<i>PerspectEvolver</i>	Literature-based	● (to the synthesis)	● (explicit)

First, collaboration reveals differences in how researchers understand and develop an initial research direction. When a researcher brings a tentative direction to colleagues, they may reach different conclusions because they rely on different assumptions, explanations, and criteria [1]. These differences can introduce alternative ways of understanding the problem and broaden the directions considered during hypothesis generation [16].

Second, researchers work through these differences through deliberation, examining where their perspectives conflict, where they align, and whether and how the developing hypothesis should change [41]. Disagreements identify questions that remain unresolved, while areas of agreement establish a shared basis for further reasoning. In this way, deliberation clarifies differences among perspectives while bringing together insights that can refine the developing hypothesis.

Third, deliberation can change the perspectives themselves. Counterarguments can qualify earlier assumptions, new evidence can change how researchers interpret the problem, and alternative explanations can become part of their perspective [5]. These revised perspectives then shape how researchers interpret subsequent information and which directions they consider next [35]. The researchers’ perspectives, therefore, develop alongside the hypotheses they explore.

2.2 Configuring and Evolving Perspectives in Multi-Agent Ideation Systems

Recent LLM-based multi-agent systems support ideation by assigning agents distinct personas that expose users to different perspectives on a problem [6, 29–31, 40]. Across these systems, agent perspectives are configured through two broad approaches: role-based personas and domain-based personas. Role-based systems differentiate agents by assigning functional, occupational, or genre roles that shape how each contributes to the task. MultiColleagues represents occupation-based personas as colleagues that interact in explore and focus modes [40]. CritiqueCrew assigns UX, product, and engineering roles to provide complementary design critiques, which a lead coordinator synthesizes and compares [6]. NarrativeLoom uses ten genre-specific personas to generate alternative story beats in parallel for users to review and select [31]. Domain-based systems differentiate agents by the research areas or expertise they represent. PersonaFlow constructs domain-expert personas from the research question, retrieved literature, or profiles of relevant authors, representing each through a role and associated expertise [30]. Perspectra selects personas from a catalog organized around research areas, focus areas, and methodological skills [29].

Table 1 compares these systems across three aspects of perspective use: how perspectives are configured, whether their contributions remain visible during deliberation, and whether the perspectives themselves evolve as the discussion develops.

Configuration divides into two approaches. Role-based systems assign agents fixed functional or genre roles—colleagues in MultiColleagues [40], UX, product, and engineering reviewers in CritiqueCrew [6], genre storytellers in NarrativeLoom [31]. Domain-based systems derive agents from research areas instead: PersonaFlow synthesizes domain-expert personas from the research question or from retrieved papers and their authors [30], and Perspectra draws agents from a catalog built by clustering a taxonomy of research areas and methodological skills [29].

In systems where agents deliberate with one another, attribution specifies the speaker of each message and thereby identifies the perspective to which that message belongs [6, 29, 31]. MultiColleagues, for instance, gives each persona a distinct profile image and role-based message styling so that contributions remain identifiable across an extended discussion; participants reported that anchoring contributions to individual colleagues made it easier to follow up on an idea and to combine ideas into coherent outcomes [?]. Attribution alone, however, requires the researcher to reconstruct how a given perspective contributed to the developing discussion: what it supported or challenged, whether its contribution was taken up or revised, and which disagreements remained unresolved.

Support for perspective evolution is more limited. In MultiColleagues, CritiqueCrew, NarrativeLoom, and PersonaFlow, agents retain their initially configured personas throughout the interaction; changing a persona requires separate user intervention [6, 30, 31, 40]. The discussion may introduce new ideas or disagreements, but the agent does not incorporate them into its perspective in later interactions. Perspectra provides partial support for this evolution through long-term memory, which periodically distills an agent’s previous contributions, retrieved literature, and challenges from other agents into research-idea snippets that inform its subsequent reasoning [29]. However, the system does not surface which parts of the deliberation motivated a particular change in the agent’s perspective.

Taken together, prior systems provide mechanisms for introducing diverse agent perspectives, but offer less support for understanding how those perspectives develop through deliberation. Attribution can identify which agent produced a response, and Perspectra allows information from prior interaction to inform later reasoning, yet the relationship between what a perspective contributes and how it subsequently changes remains largely implicit. Researchers therefore have limited support for understanding how a perspective is configured, how it contributes during deliberation, and whether and how it changes through that deliberation [38].

PerspectEvolver addresses this gap by making the development of Perspectives inspectable across formation, deliberation, and evolution. *Perspective Formation* forms literature-grounded Perspectives so researchers can examine what informs each interpretation. *Perspective-Guided Deliberation* makes each Perspective’s contributions visible during deliberation and shows how those contributions affect the developing discussion. *Perspective Evolution* makes subsequent Perspective changes inspectable by showing what changed, why it changed, and which discussion prompted it. Together, these phases support researchers in understanding how Perspectives are formed, how they contribute during deliberation, and how they develop through that deliberation.

2.3 Divergent and Convergent Thinking Activities in Research Ideation

Scientific hypothesis generation is an iterative process of reasoning, exploring a vast hypothesis space and refining hypothesis candidates [39, 43]. **Divergent thinking** widens the region of the hypothesis space a researcher explores, pushing exploration past what they already know; Schunn and Klahr [44] observe that exhaustively searching available datasets surfaces insightful features precisely because it works through scientific facts thoroughly. **Convergent thinking** then narrows that space, weighing candidates against multiple criteria, and Dunbar [10] argues that verifying and focusing on particular data types or characteristics is indispensable to scientific discovery. Iteration of both modes, repeating cycles of divergence and convergence, iteratively filters the expanded space until a small set of dependable

candidates remains. As Sawyer [42] emphasizes, a creative idea does not develop linearly within either mode, but through a non-linear process in which the two alternate again and again. Supporting both modes, along with the movement between them, is what enables researchers to arrive at hypotheses they can hold with confidence [24, 37, 43, 44].

Researchers’ work in this alternation divides into two kinds, which we call *advancing* and *grounding*. Advancing changes the state of the working hypothesis: a candidate is introduced, two are merged, or one is kept and the rest set aside. Grounding leaves that state intact and brings existing material to bear on it—prior work, established criteria, or the researcher’s own knowledge. Both are essential for developing promising hypotheses, but they contribute differently: grounding shapes how candidates are judged, while advancing is what changes them. Introducing, merging, and committing to candidates each leave the working hypothesis in a different state, and the next round of deliberation builds on that updated state [42]. Through this repeated updating, the region of the hypothesis space a researcher covers widens, and a promising hypothesis emerges within that region [16, 44].

In this work, we explore designs that nudge participants to engage in divergent and convergent thinking through sensemaking over perspectives and deliberation. More specifically, we design perspective-guided deliberation in which users interact with diverse perspectives through activities such as synthesis, integration, reflection, and exploration, so that the interaction itself provides grounding and researchers can focus on advancing the working hypothesis.

3 Formative study

The formative study aimed to understand (1) how researchers collaborate with colleagues to develop scientific hypotheses; (2) how researchers turn to AI as an alternative when collaboration with colleagues is not feasible; and (3) the limitations and opportunities of existing AI systems for supporting hypothesis development, which informed our design goals for a multi-agent system.

3.1 Participants

We recruited six participants: three postdoctoral researchers and three PhD candidates from bioengineering, electrical and computer engineering, mechanical engineering, and chemical engineering (Table 2). Participants had 4–8 years of research experience ($M = 5.5$, $SD = 1.87$). We recruited participants through personal referrals and outreach to research labs. The Institutional Review Board (IRB) approved the study, and all participants provided informed consent. Each participant received a \$30 gift certificate. When reporting interview quotes, we use P_n to denote the participant number.

3.2 Study Procedure

The study consisted of three parts. First, we conducted a 30-minute in-depth interview covering (1) participants’ workflow in collaboration for hypothesis ideation, (2) their goals and strategies for using AI tools when collaborating with colleagues is not feasible, and (3) the challenges and limitations they encounter when using AI tools for hypothesis development. Because participants had no prior experience with multi-agent systems, we focused on their use of existing AI tools. Next, we observed participants for 20 minutes as they developed a hypothesis for a research problem from their ongoing work. Participants used their usual general-purpose LLM service (ChatGPT for all six) as they normally would, prompting it to adopt perspectives relevant to the research problem. Finally, a 30-minute post-task interview examined participants’ refinement strategies, challenges, and desired support. Each session lasted 1.5 hours. We analyzed the interview transcripts using thematic analysis and iterative open coding [8]. Two authors coded the transcripts for emerging themes, which the co-authors discussed until reaching consensus.

3.3 Findings

3.3.1 Collaboration for Hypothesis Generation. We found that collaborative hypothesis development involved three recurring stages: (1) formulating an initial hypothesis from domain literature, (2) engaging with collaborators who held different perspectives, and (3) evolving perspectives and refining hypotheses around unresolved issues. First, researchers formulated an initial hypothesis by interpreting domain literature through their own perspective. They reviewed prior work to identify findings, mechanisms, and unexplored areas, then organized them around questions aligned with their interests and perspectives. As P_6 described, “*I start with papers that align with my interests and perspectives, particularly work by researchers whose perspectives are similar to my own. I also look through papers that I have already identified or listed, tracing what has been tried and what has not, until the hypothesis feels sharp enough.*” Next, researchers engaged with collaborators who held different perspectives. These collaborators introduced alternative viewpoints that revealed blind spots in the researchers’ initial perspectives. This prompts researchers to reconsider and expand their perspectives. As P_1 described “*During our meeting, other researchers challenged my perspective on the differentiation timeline, asking if the early-stage cell population was too heterogeneous. That feedback made me reconsider my assumptions about the timing of differentiation and scrutinize our flow cytometry data from scratch.*” Finally, researchers used unresolved issues from earlier discussions to initiate further rounds, evolving their perspectives and refining their hypotheses through continued collaboration. For example, P_3 described “*After a few months working with the control engineers, their pushback on the hardware side stopped feeling like a roadblock. I started reading it as a constraint I just hadn’t accounted for yet. Once I could think in their terms too, the hypothesis got a lot sharper.*”

This collaborative process typically unfolds over several weeks to months before experimental validation. However, collaboration faced practical barriers from limited professional networks, scheduling constraints, and discomfort receiving critical feedback. For example, “*At first, I found their questions difficult because they felt like challenges to myself. It was uncomfortable to share an idea that was still incomplete.*” (P_5). “*The collaborators I can work with are largely limited by my own network. Also, it is often difficult to meet regularly because of our individual schedules.*” (P_2). As a result, researchers used AI as an accessible alternative for exploring and refining hypotheses.

3.3.2 Roles of AI in Research Ideation. Although prior multi-agent systems support diverse agent perspectives, none of our participants had prior experience with such systems. Instead, they relied on general-purpose AI services such as ChatGPT¹ for individual hypothesis ideation. To simulate collaboration, participants constructed alternative perspectives, deliberated with them, and updated them as discussions developed. First, participants constructed an alternative perspective by uploading papers with views different from or opposing their own and providing instructions to contextualize that perspective.

As P_4 described, “*When I was thinking about how to improve AI accuracy, I uploaded papers that focused more on efficiency, because that was a somewhat different perspective from my own. I wanted the LLM to help me compare these perspectives and find where they could be connected.*”

Second, participants deliberated with these perspectives, either introducing them into an existing conversation or using separate conversations to keep their responses independent. Participants then summarized the discussions and compared their outputs to identify converging and conflicting points. As P_5 described, “*Sometimes I bring another perspective into the same conversation, but when I want to keep the opinions separate, I start a new chat. Then I look at the summary or at what each perspective said, and work out how to develop my hypothesis from there.*” Participants also reconsidered how they framed the research problem in response to different perspectives. As P_2 described, “*I was*

¹<https://chatgpt.com/>

initially thinking about toxicity as a problem with the compound itself, but the AI pointed out that the toxicity could depend on a specific metabolic pathway or cell type. That made me reconsider whether to focus on the compound or the underlying biological pathway.”

Finally, participants refined their hypotheses and updated the AI perspectives based on the discussion before continuing to further rounds. They incorporated convincing feedback into the hypothesis, while other feedback prompted them to search related literature or reframe the problem before updating the perspective and continuing. As P_6 described, “When I searched for ways to improve the drug’s solubility, the AI suggested polar groups with very different structures from my original compound, which made me reconsider the shape of the drug. When I was not fully convinced by something in the conversation, I looked for papers on similar structural modifications and used them to update the AI’s perspective so we could have a more productive discussion.”

3.3.3 Key Challenges in the Current Workflow. Our workflow observations revealed three challenges in configuring, deliberating with, and updating AI perspectives.

First, participants had no structured way to specify how an agent should interpret uploaded papers, leaving its stance, assumptions, and evidentiary standards opaque and sometimes misaligned with their intent. As P_1 described, “I gave it three of Baker’s papers and said ‘think like this,’ but I couldn’t tell what it actually took from them. It just sounded like a generic biologist.” P_2 described a related difficulty distinguishing multiple perspectives: “I would set up several perspectives, but it was hard to tell clearly what made them different. The more the discussion went on, the stronger that ambiguity got.”

Second, participants had difficulty understanding how each perspective shaped the discussion. Even when the discussion reached an agreement or exposed a disagreement, they could not tell what each perspective had contributed to that result. As P_2 described, “Using several agents was useful, but it was hard to know what each of them had contributed to a given agreement.” This made it harder to judge how the discussion had arrived at its conclusions and which perspective to involve next. P_6 added, “An agreement would come out, but I couldn’t tell whether one side’s position had been taken up or whether the two had met somewhere. So nothing came to mind about which perspective to add next.”

Third, participants had difficulty determining how discussion outcomes should revise each perspective. They summarized the discussion, interpreted its implications themselves, and manually updated the corresponding perspective prompts. As P_3 described, “The agents sometimes drifted away from the new discussion and its context, so I updated their perspectives myself. If I didn’t, the context seemed to get tangled later on.” These revisions were not tracked as part of the discussion, leaving participants to remember what had changed and why across rounds. P_4 also described uncertainty about how to translate discussion insights into perspective updates: “Even when I pull an insight out of the deliberation, I’m not sure how to reflect it. If I miss it, a gap seems to open between me and the perspectives later, so I go back and update them again.”

3.3.4 Elements of a Perspective. Our findings showed that participants often defined AI perspectives indirectly through papers and instructions, making it difficult to understand what distinguished one perspective from another. We therefore asked participants to examine their own research papers and identify the elements that distinguished different perspectives. Drawing on prior work on framing [11], we used two higher-level concepts as a starting point: *Framing*, how a perspective understands the research problem, and *Position*, where it stands within that framing. From these interviews, we identified four foundational elements, which we represent as *Perspective Fragments: Scope, Explanation, Approach*, and *Significance*. These fragments characterize different aspects of a perspective and can inform both its Framing and Position.

- *Scope*: what a perspective covers, including the phenomena, populations, settings, and conditions under consideration.
- *Explanation*: how a perspective accounts for the phenomenon through relevant relationships, mechanisms, and processes.
- *Approach*: how a perspective investigates the problem through measurement, comparison, testing, or evaluation.
- *Significance*: why the problem matters, including the consequences or implications that make it worth investigating.

3.4 Design Goals

Based on these findings, we proposed three design goals for our system:

- **DG1. Representing a perspective as inspectable elements.** When configuring perspectives for multiple agents, the interface should make explicit what each perspective takes from the literature, so that users can tell one perspective from another.
- **DG2. Surfacing each perspective’s contribution to the deliberation.** When several perspectives deliberate, the interface should record what each perspective contributed across the deliberations, so that users can judge how far those outcomes can be trusted without rereading the exchanges.
- **DG3. Carrying what the deliberation settled into the perspective evolution and the hypothesis.** When a deliberation concludes, the interface should make explicit how its outcome revises each perspective and the hypothesis, and leave what remains open as the starting point for the round that follows.

4 System and Implementation

Building on these design goals (DG1–3), we present *PerspectEvolver*, an interactive multi-agent system for research ideation through deliberation among diverse Perspectives. Each Perspective is grounded in an area of scientific literature and represents how research in that area frames, explains, and investigates the research problem. Researchers select Perspectives to form a Perspective Ensemble. Within the Ensemble, *PerspectEvolver* externalizes each Perspective as a Perspective Agent that contributes its interpretation and supporting evidence during deliberation.

PerspectEvolver organizes this process into three phases. *Perspective Formation* supports researchers in exploring the literature and forming an initial Perspective Ensemble. *Perspective-Guided Deliberation* develops research directions through Proposals, Deliberative Threads, and Working Syntheses. *Perspective Evolution* refines affected Perspectives, proposes Investigation Brief revisions, and generates follow-up questions from synthesized discussions. Additional technical details appear in Appendix A.4.

4.1 Perspective Formation: Exploring and Forming Perspectives (DG1)

Perspective Formation supports researchers in exploring alternative interpretations of a research problem by forming Perspectives grounded in relevant literature (Figure 2). Elicitation establishes the Investigation Brief, Discovery retrieves and organizes relevant literature, and Formation constructs candidate Perspectives for researchers to assemble into a Perspective Ensemble.

4.1.1 Elicitation. Researchers begin by defining an Investigation Brief comprising the Problem, Framing, Previous Work, Methodology, and Expected Results (Figure 2-①). The brief remains visible and editable throughout the investigation (Figure 4, left), recording the researcher’s problem framing and intended study.

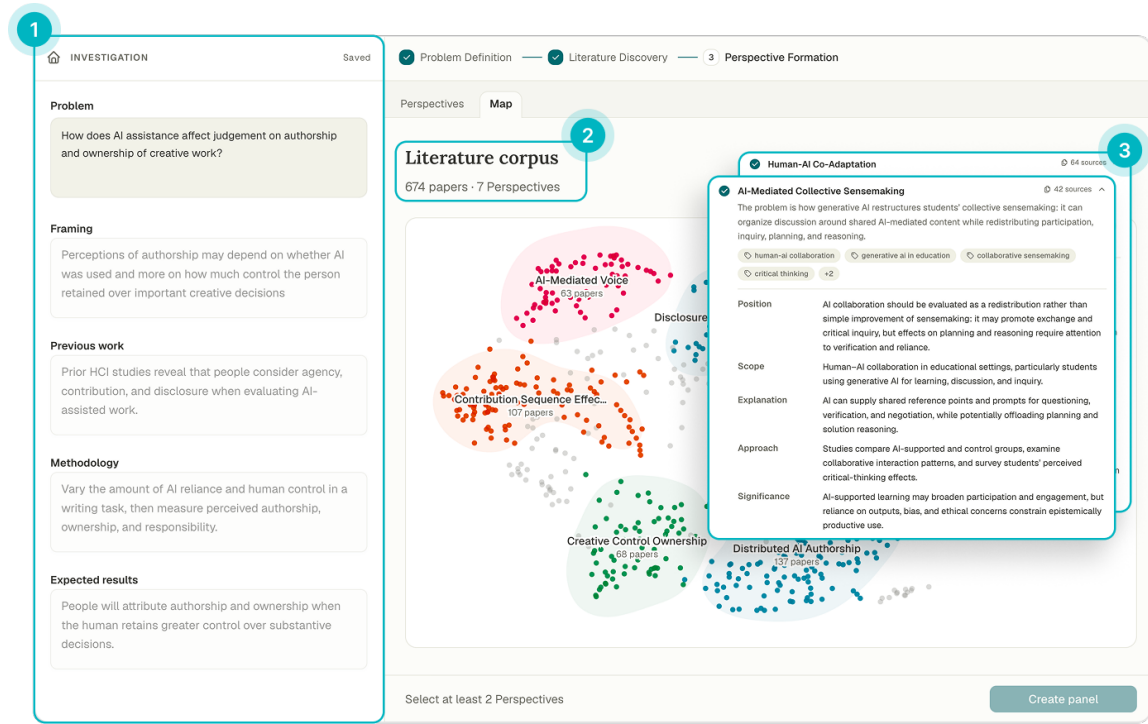


Fig. 2. *Perspective Formation* interface. ① The Investigation Brief records the Problem, Framing, Previous Work, Methodology, and Expected Results and remains editable throughout the investigation. ② The Literature Map presents the retrieved corpus (674 papers) within labeled literature regions from which candidate Perspectives (seven) are formed. ③ Perspective Cards present each Perspective's Framing, Position, Scope, Explanation, Approach, and Significance with supporting sources. Researchers select two or three Perspectives and choose *Create panel* to form the Perspective Ensemble.

4.1.2 Discovery. From the Investigation Brief, *PerspectEvolver* generates research directions and translates them into queries for Semantic Scholar's snippet search [23]. The search returns relevance-ranked passages from indexed titles, abstracts, and body text, allowing retrieval through findings, methods, or arguments not prominent in titles or abstracts. Retrieved passages can prompt additional searches for topics not covered by the initial queries.

The system obtains metadata and SPECTER2 embeddings [46] for retrieved papers and organizes them into literature regions using UMAP [33] for dimensionality reduction and HDBSCAN [32] for density-based clustering. The Literature Map presents papers within these labeled regions for researchers to inspect and compare (Figure 2-②). Papers in each region provide source material for a candidate Perspective.

4.1.3 Formation. From each literature region, *PerspectEvolver* forms a candidate Perspective representing one interpretation of the research problem. Each Perspective includes a Framing that describes how it understands the problem and a Position that expresses its conclusions, qualifications, uncertainties, or proposed directions. Perspective Fragments characterize its *Scope*, *Explanation*, *Approach*, and *Significance*. Each field references the Literature Evidence used to form it, with each evidence item linked to its source passage.

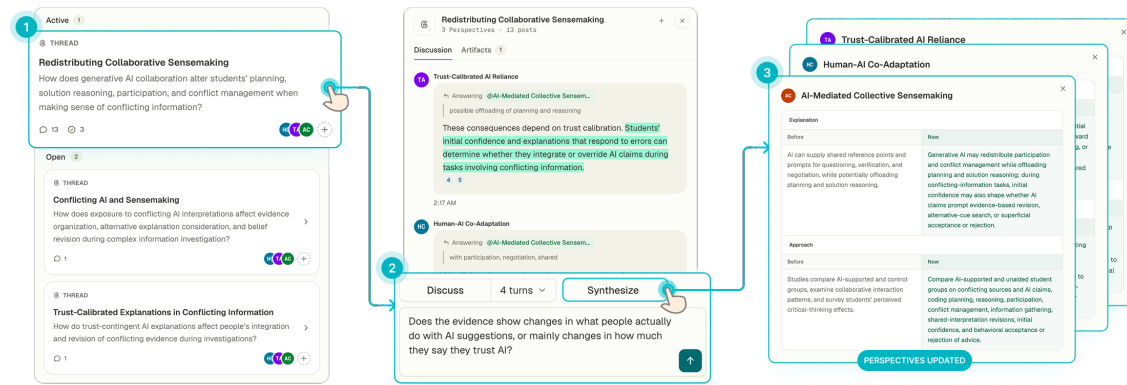


Fig. 3. Perspective-guided deliberation and the transition to Perspective Evolution. ① Deliberative Threads are organized as *Active* or *Open*, with each card showing its research question, activity, and participating Perspectives. ② Selecting a thread opens its discussion, where contributions identify the contributing Perspective and what they address. Researchers request additional turns with *Discuss* or select *Synthesize* to produce a Working Synthesis. ③ Perspective Reflection presents affected fields by comparing their previous and revised content.

Candidate Perspectives appear as Perspective Cards alongside the Literature Map (Figure 2-③). Researchers compare the candidate interpretations and inspect their supporting sources before selecting two or three to form a Perspective Ensemble. Choosing *Create panel* begins deliberation with the selected Perspective Agents.

4.2 Perspective-Guided Deliberation: Thread-based Deliberation (DG2)

Perspective-Guided Deliberation develops research directions through Proposals, Deliberative Threads, and Working Syntheses (Figure 3-①–②). Proposals introduce directions for inquiry, Deliberative Threads develop them through discussion among Perspective Agents, and Working Syntheses record discussion outcomes at points chosen by the researcher.

4.2.1 Proposals. Each selected Perspective introduces a Proposal comprising a research question, tentative hypothesis, and rationale. Researchers *Keep* or *Skip* each Proposal and can *Undo* either decision. Confirming retained Proposals creates a Deliberative Thread for each direction.

Each thread initially includes the selected Perspectives so that a Proposal from one can be examined through the others' interpretations and literature. Researchers can also introduce their own research question and choose which Perspectives participate.

4.2.2 Deliberative Threads. Each Deliberative Thread organizes discussion around one research question. Perspective Agents contribute interpretations and supporting evidence from their respective Perspectives. One Perspective may challenge an explanation with contrary evidence, qualify it by introducing a relevant condition, or offer an alternative explanation.

Threads are organized as *Active* or *Open*, with each card showing its research question, discussion activity, and participating Perspectives (Figure 3-①). The Canvas presents threads as branches from the investigation's research question, allowing researchers to move among research questions while seeing how their threads relate (Figure 4, right). Active threads can continue to develop, while Open threads remain available for later investigation.

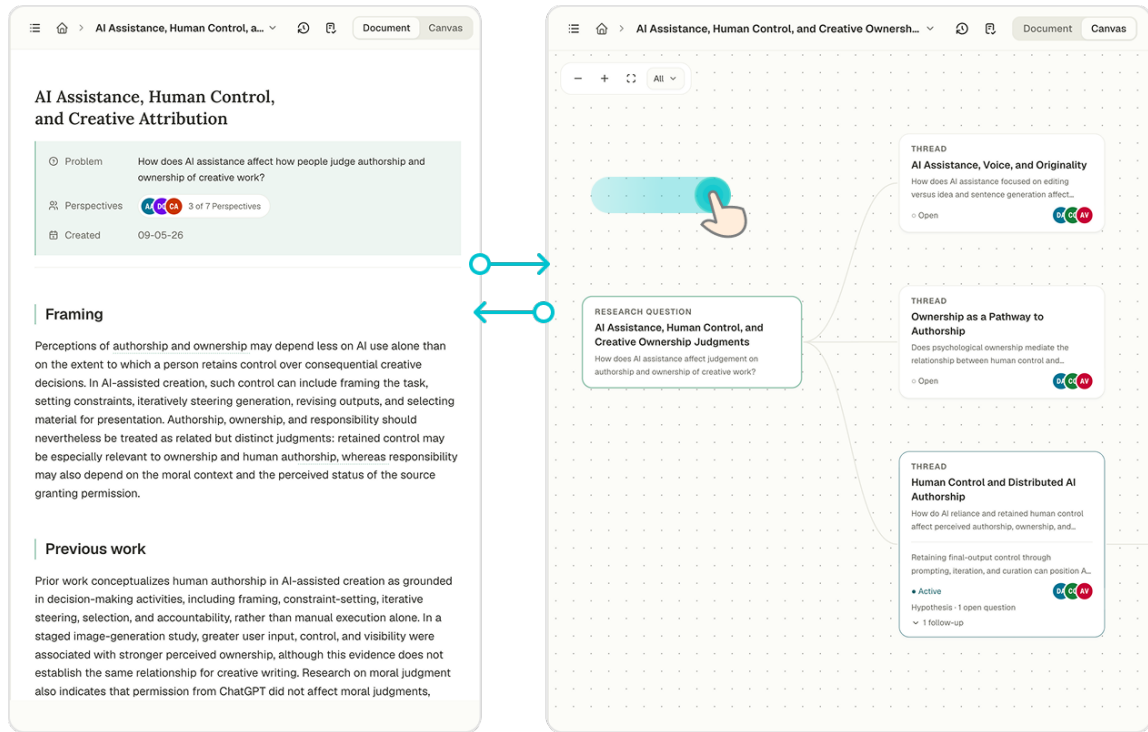


Fig. 4. Document and Canvas views of an investigation. The Document view (left) presents the Investigation Brief with the research problem and participating Perspectives. The Canvas view (right) organizes Deliberative Threads as branches from the research question. Synthesized threads can expose resulting hypotheses, open questions, and follow-up directions, while suggestions appear as prospective branches before researchers decide whether to pursue them.

Selecting a thread opens its discussion. Each contribution identifies its Perspective and the earlier contribution or passage it addresses (Figure 3-②). Empirical claims include numbered literature references so researchers can distinguish a Perspective's interpretation from its supporting evidence.

Discuss requests one to six additional turns, while *Stop* interrupts the exchange. Researchers can reply to or @-mention a Perspective and add another Perspective when its interpretation becomes relevant. Selecting text focuses the next interaction on that passage. *Quote response* responds directly to its author, *Challenge* requests supporting evidence, and *Expand* asks for further explanation.

4.2.3 Working Synthesis. Selecting *Synthesize* produces a Working Synthesis of the discussion at that point. It consolidates findings, limitations, unresolved disagreements, and open questions. Compatible ideas are integrated while unresolved differences remain explicit, and empirical findings retain links to supporting literature. When the discussion motivates a testable hypothesis, the synthesis also records a tentative prediction and rationale.

The thread's *Artifacts* view stores a *Prediction*, its *Limitations*, and an *Evaluation Plan*. On the Canvas, a synthesized thread can expose its hypothesis, open questions, and follow-up directions within the investigation structure (Figure 4, right). Researchers can return to a synthesized thread, question or extend its conclusions, and synthesize it again.

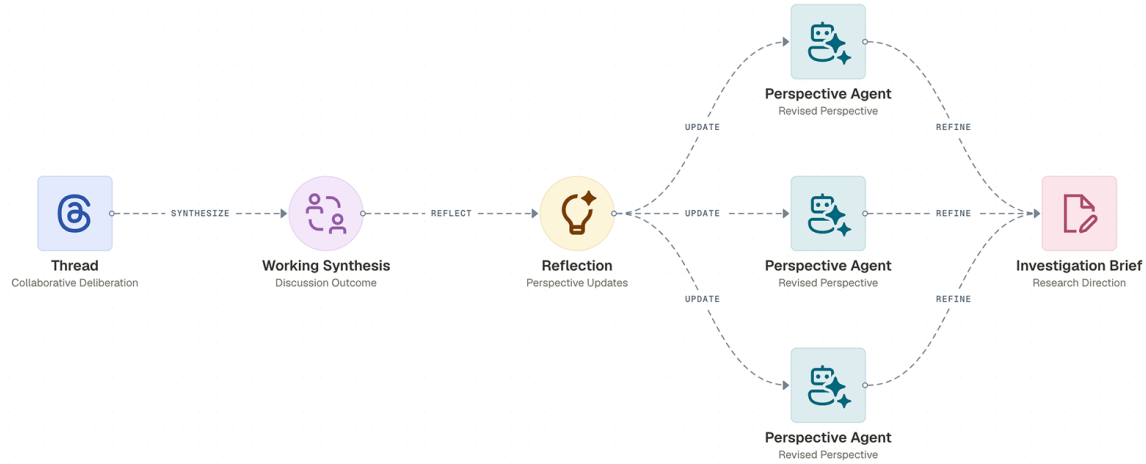


Fig. 5. *Perspective Evolution* workflow. A Deliberative Thread produces a Working Synthesis of the discussion. Reflection compares participating Perspectives with provenance-linked inputs extracted from that discussion and refines affected content. The Working Synthesis also informs proposed Investigation Brief refinements and follow-up questions.

After synthesis, *PerspectEvolver* examines whether the discussion requires refinement of participating Perspectives (Figure 3-③).

4.3 *Perspective Evolution: Reflection, Refinement, and Emergent Threads* (DG3)

Following synthesis, *PerspectEvolver* refines affected Perspectives, proposes Investigation Brief changes, and generates follow-up questions. Reflection evaluates whether Discussion Inputs require changes to participating Perspectives, while document refinement compares the Working Synthesis with the Investigation Brief (Figure 5). Unresolved questions can become new Deliberative Threads for subsequent investigation.

4.3.1 *Perspective Reflection.* After synthesis, *PerspectEvolver* extracts Discussion Inputs that distinguish empirical findings from scientific considerations and retain references to their supporting evidence and discussion sources. Reflection compares these inputs with each participating Perspective and refines it only when an input contradicts or qualifies an existing statement or resolves an expressed uncertainty.

Reflection is constrained to change only affected Perspective fields. When a Perspective changes, revision cards compare its previous and revised content by field (Figure 3-③). Researchers can inspect how the discussion refined a Perspective rather than seeing only its current state. *History* records each revision with its previous and revised text, *Why this changed*, and the Deliberative Thread that prompted it.

4.3.2 *Document Refinement.* From the same Working Synthesis, *PerspectEvolver* proposes revisions to affected sections of the Investigation Brief (Figure 5). Each proposal shows additions and deletions within the affected section together with the reason for the revision.

Researchers decide which changes enter the Investigation Brief through *Accept*, *Edit*, or *Reject*, then confirm them with *Apply decisions*. This keeps changes to the research direction under researcher control. Researchers can also revise the brief directly. *Annotate* attaches a note to selected text or lets the researcher @-mention a Perspective, while *Rewrite*

previews a proposed replacement before application. Direct editing remains available throughout. The Document view presents the revised Investigation Brief (Figure 4, left), which subsequent deliberation uses as context.

4.3.3 Emergent Threads. *PerspectEvolver* uses the Working Synthesis and current Investigation Brief to propose follow-up questions from unresolved disagreements, limitations, and open questions. Suggestions appear under *Suggested Threads* and as prospective Canvas branches (Figure 4, right) for researchers to review before opening them. Selecting a suggestion opens a new Deliberative Thread linked to its originating discussion. Subsequent threads use the latest versions of participating Perspectives.

4.4 Implementation

PerspectEvolver is implemented as a web application with a Next.js², React, and TypeScript frontend and a Python backend. React Flow³ and Dagre⁴ implement the Canvas, D3⁵ renders the Literature Map, and Tiptap⁶ supports document editing and annotations. FastAPI⁷ exposes backend endpoints, PostgreSQL⁸ stores investigation and revision state, and LangGraph⁹ coordinates checkpointed human-in-the-loop workflows.

DSPy [22] implements structured language-model operations. Perspective Agents share task-specific modules, which receive the current Perspective, evidence, and discussion context as inputs. We use OpenAI’s gpt-5.6-luna and gpt-5.6-terra models¹⁰. gpt-5.6-luna supports search planning, evidence extraction, Perspective naming, proposal generation, and interactive deliberation, while gpt-5.6-terra supports Perspective synthesis, Working Synthesis, Reflection, document revision, and follow-up generation. Literature retrieval follows the Discovery pipeline described in §4.1.2. Voyage rerankers¹¹ select evidence during Formation and deliberation. Evidence extraction identifies scientific claims and study details from retrieved passages and retains only items whose supporting quotations occur verbatim in their mapped source passages. Perspective validation requires Literature Evidence references for every nonempty field. Perspective Reflection compares Discussion Inputs with each Perspective and proposes refinements only to fields it identifies as affected. Additional model, retrieval, agent, and workflow details appear in Appendix A.4.

5 User study

We conducted a within-subjects study comparing *PerspectEvolver* with a group-chat baseline. We examined hypothesis generation, system use, and divergent and convergent thinking through three research questions.

- **RQ1:** How does *PerspectEvolver* affect hypothesis quality and researchers’ cognitive load during hypothesis generation?
- **RQ2:** How do researchers use *PerspectEvolver*’s features over a session?
- **RQ3:** How does *PerspectEvolver* shape divergent and convergent thinking activities?

²<https://nextjs.org/>

³<https://reactflow.dev/>

⁴<https://github.com/dagrejs/dagre>

⁵<https://d3js.org/>

⁶<https://tiptap.dev/>

⁷<https://fastapi.tiangolo.com/>

⁸<https://www.postgresql.org/>

⁹<https://www.langchain.com/langgraph>

¹⁰<https://developers.openai.com/api/docs/models>

¹¹<https://docs.voyageai.com/docs/reranker>

Table 2. Participant demographics and involvement. YoE: years of research experience. F: formative study; U: user study; C: prompt coding; R: hypothesis rating.

ID	Major / Discipline	Profession	YoE	F	U	C	R
P_1	Bioengineering	Postdoctoral Researcher	7	✓		✓	
P_2	Bioengineering	Postdoctoral Researcher	6	✓	✓		
P_3	Electrical and Computer Engineering	PhD Student	4	✓	✓		
P_4	Electrical and Computer Engineering	PhD Student	5	✓		✓	✓
P_5	Mechanical Engineering	PhD Student	4	✓			✓
P_6	Chemical Engineering	Postdoctoral Researcher	8	✓			
P_7	Computer Science	Undergraduate Student	2		✓		
P_8	Health Sciences	PhD Student	3		✓		
P_9	Electrical and Computer Engineering	Master's Student	3		✓		
P_{10}	Health Sciences	Master's Student	2		✓		
P_{11}	Computer Science	Master's Student	2		✓		
P_{12}	Mechanical Engineering	PhD Student	5		✓		
P_{13}	Mechanical Engineering	PhD Student	5		✓		
P_{14}	Computer Science	PhD Student	4		✓		

5.1 Baseline

Based on our formative study, in which researchers used mainstream chat-based LLMs for hypothesis development, we implemented a group-chat MAS baseline that preserved the core workflow of literature retrieval, multi-agent deliberation, and hypothesis revision while omitting *PerspectEvolver*'s explicit perspective-sensemaking mechanisms.

Perspective Formation retained query recommendation and paper retrieval, but represented Perspectives through simpler paper-grounded personas rather than literature clustering and structured Perspective extraction. *Perspective-Guided Deliberation* retained multi-agent deliberation, suggested topics, and discussion summaries, but did not explicitly surface how each Perspective contributed to the discussion. The document-refinement workflow retained the hypothesis document and transfer of discussion outcomes, but did not support explicit Perspective evolution or follow-up question generation.

5.2 Study Procedures

We conducted a within-subjects study with 10 participants to compare *PerspectEvolver* and a baseline group-chat MAS. The study evaluated how the different interaction designs affected research ideation outcomes and perceived cognitive load (RQ1), feature use over a session (RQ2), and divergent and convergent thinking activities (RQ3).

5.2.1 Participants. We recruited 10 participants (Table 2), including engineering and science undergraduates, master's students, PhD students, and postdoctoral researchers with research experience ranging from 2 to 6 years ($M = 3.60$, $SD = 1.43$). Participants had diverse domain backgrounds and were recruited through university mailing lists and social media platforms. The study was conducted remotely over Zoom¹² and participants were compensated with \$40. This study was approved by the university's Institutional Review Board (IRB).

¹²<https://www.zoom.com/>

5.2.2 Study Design and Procedure. We used a counterbalanced within-subjects design in which each participant completed two hypothesis-generation tasks: one using *PerspectEvolver* and one using the baseline group-chat interface. Participants submitted a scientific hypothesis on their research problem as the final output, consisting of four sections: motivation, previous work, methodology, and expected results. Before the study, participants completed a pre-session questionnaire covering demographics, research background, and frequency of LLM use. The questionnaire also collected two research problems, one for each condition, which participants had prepared to meet two criteria: interest and familiarity, in that the problem concerned a topic they were interested in and sufficiently familiar with; and comparable difficulty, in that the two problems were similar in scope and complexity and demanded a similar level of effort and domain knowledge.

Each session lasted approximately 120 minutes: a five-minute introduction covering the study goals and consent procedures; two 50-minute tasks separated by a five-minute break; and a 10-minute semi-structured interview. Each task lasted 50 minutes and comprised four phases. First, participants completed a five-minute walk-through of the corresponding interface, in which the researcher demonstrated every feature using a sample research idea, and participants practiced using the system. Second, participants completed a 10-minute solo phase in which they drafted the four sections of a hypothesis while consulting papers on Google Scholar; this draft served as the point of comparison for the final output. Third, 30 minutes interacting with the system. Fourth, a five-minute post-session survey covering cognitive load, usability, and divergent and convergent thinking support, all on seven-point Likert scales. The session concluded with a 10-minute semi-structured interview on participants' experience with both systems, changes in their scientific reasoning, the experience of configuring and evolving perspectives during hypothesis generation, divergent and convergent thinking, and which features they found most helpful for scientific ideation (§A.2 in the appendix). Throughout both tasks, participants were instructed to think aloud, verbalizing their thoughts, reactions, and decisions, and we recorded these sessions for later analysis.

5.3 Measurement

To evaluate our research questions, we collected and analyzed qualitative and quantitative data, including improved hypothesis quality, surveys, think-aloud data, and interviews.

5.3.1 Quality of Submitted Hypotheses. We performed an LLM-as-a-judge evaluation of the quality of participants' hypotheses. We used OpenAI's GPT-5.6 Luna¹³ to assess quality on the same criteria as the self-assessment survey: novelty, plausibility, significance, testability, and evidence support. We supplied both the initial hypothesis and the revised final hypothesis as context, and prompted the model to rate each on a 1–7 Likert scale across all five dimensions (§A.3 in the appendix) in a pairwise manner. We then took the difference between the two scores of the final and the initial hypotheses as the degree of improvement, where the delta for each metric m and pair i is $\Delta_{i,m} = \mu_{i,m}^{\text{final}} - \mu_{i,m}^{\text{init}}$. To ensure consistency, we ran ten independent judgments and averaged the scores by dimension for each participant and condition. To establish the reliability of the LLM-as-a-judge ratings, we sampled ten hypothesis pairs (the five rated highest and the five rated lowest by the LLM) and had two PhD candidate researchers in electrical and computer engineering (P_4) and mechanical engineering (P_5), who participated in only the formative study, rate them independently using the same rubric. Across the five dimensions, we obtained an average Krippendorff's alpha of 0.75 with the LLM, indicating substantial agreement.

¹³<https://developers.openai.com/api/docs/models/gpt-5.6-luna>

Table 3. Codebook for participants' prompts. Eight subprocesses adapted from Boldt [2] and Boldt and Kaufman [3].

Code	Definition	Example
GENERATION	Proposes a new idea that builds on the discussion so far	"Does annealing-induced interlayer crystallization improve electronic transport by simultaneously reducing residual stress and eliminating interfacial structural defects in GaN-on-Si?"
EXPLORATION	Pushes the discussion toward a new direction outside what has been discussed	"SO, the user satisfaction has no impact, which means, researchers had to look into cognitive load? What does that mean for this question?"
SYNTHESIS	Combines two or more existing candidates into one and carries it forward	"The diffusion-kinetics account and the baseline-stability account both explain the 37 °C signal advantage—is there something that satisfies both?"
SELECTION	Commits to one candidate and sets the rest aside	"Between user satisfaction and cognitive load, go with cognitive load and narrow the hypothesis to that."
COMPARISON	Brings in a new reference, methods, or cases to check the current discussion against	"Okay, is that more effective than the clinic based monitoring?"
KNOWLEDGE	Draws on prior knowledge or experience as grounds for the current discussion	"In most of the competitive ELISA, the nonspecific binding is always the highest signal which could concentrate, so it is somehow impossible to increase the binding."
EVALUATION	Judges the validity, strength, or fit of a candidate already present	"then any limitations for applying home monitoring for the patients?"
ANCHORING	Revisits the goals and constraints of the task, tightening the criteria the discussion rests on	"keep in mind we are building to specifically simulate a scientist not just any human. A domain expert who can on behalf of human interact with a H-AI or HCI system"

5.3.2 Analysis of System Logs and Prompts. We collected every prompt participants sent to the agents in both conditions and analyzed them qualitatively for moments of divergent and convergent thinking (Table 3). We first compiled the full set of creative subprocess codes from two prior works [2, 3], yielding 13 codes (§A.1 in the appendix). Two researchers, one in electrical and computer engineering (P_4) and one in bioengineering (P_1), who participated only in the formative study, piloted this set on a subset of the prompts. We merged rarely used codes into related ones or dropped them, producing the final set of eight codes reported in Table 3. The same two researchers then coded all prompts independently with the final codebook, reaching a Cohen's κ of 0.82. Remaining discrepancies were resolved through discussion. After excluding prompts that could not be assigned to any code (e.g., vague or off-task responses), 23 prompts were excluded from the original 162 prompts (83 Baseline, 79 *PerspectEvolver*), leaving 55 Baseline and 61 *PerspectEvolver* instances (116 total) that were coded using the eight categories.

5.3.3 Survey. Post-session surveys collected participants' subjective feedback on their experience: the NASA-TLX [14] for cognitive load, the System Usability Scale (SUS) [4] for usability, and questions on support for divergent and convergent thinking [20]. The full set of survey items is provided in Table 4 in the appendix.

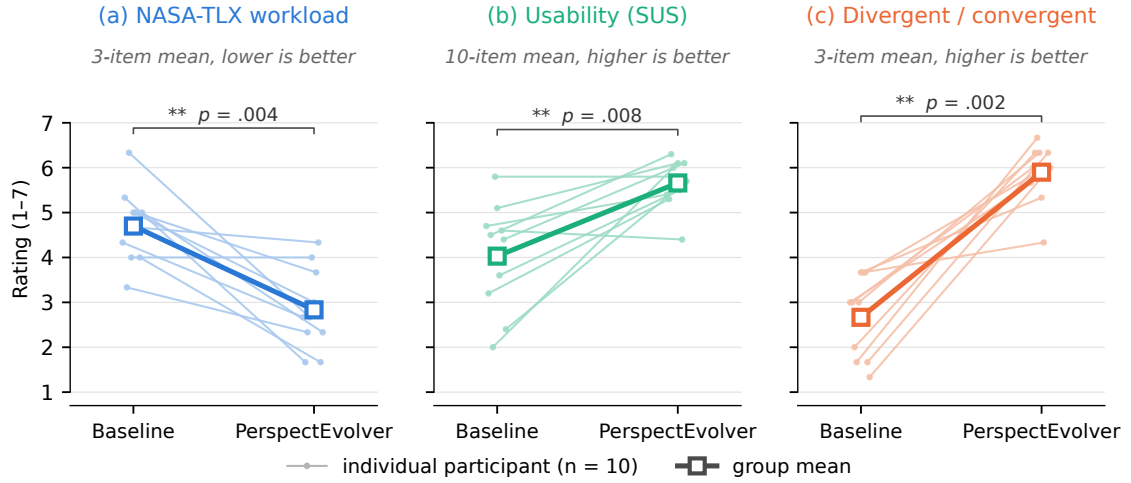


Fig. 7. Line plots showing workload, usability, and divergent/convergent support by condition ($n = 10$, within-subjects). Participants rated (a) NASA-TLX workload, (b) usability, and (c) divergent/convergent support for both the baseline and PerspectEvolver using a 7-point Likert scale. Lower is better in (a) and higher in (b) and (c). Light lines show individual participants; bold lines show group means. Wilcoxon signed-rank exact (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

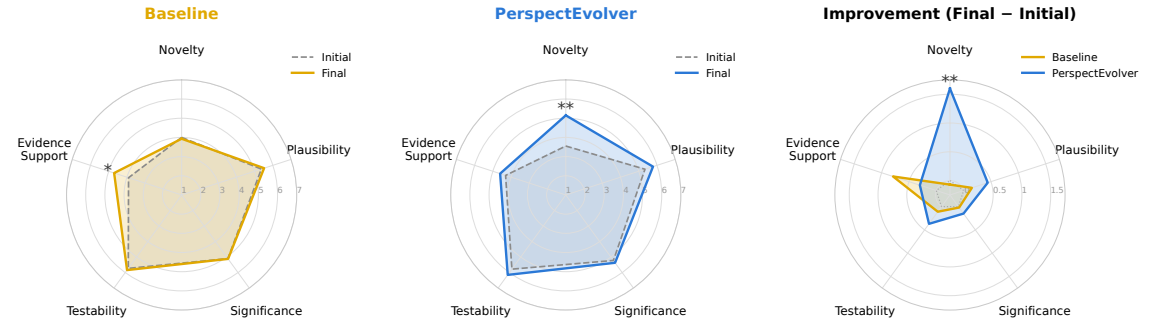


Fig. 6. Radar graphs of hypothesis quality across five dimensions for the Baseline and PerspectEvolver conditions (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Initial and final ratings (1–7) per condition (left, center) and improvement between them (right).

6 Results

We report findings organized by our research questions (RQ1-3), analyzing user interactions, survey responses, and think-aloud data results from the user study. All comparisons use two-sided Wilcoxon signed-rank exact tests [47] on within-participant pairs, with matched-pairs rank-biserial correlation as the effect size.

6.1 PerspectEvolver Enhances Hypothesis Quality without Increasing Cognitive Load (RQ1)

In this section, we present improvements in users' hypothesis edits (§6.1.1), the user-perceived usability and cognitive load across each condition (§6.1.2).

6.1.1 *Improved Novelty of Written Hypotheses.* Figure 6 shows how hypothesis quality changed from the initial draft to the final version under each condition, as rated by an LLM judge on five dimensions (§5.3.1). We compare the initial-to-final change in each arm, paired across participants.

Novelty improved substantially more under *PerspectEvolver* than under the baseline ($M = +1.60$ vs. -0.07 , difference $+1.67$, $p = .002^{**}$), with all ten participants moving in the same direction. Plausibility ($M = +0.44$ vs. $+0.15$, $p = .641$), Significance ($M = +0.15$ vs. $+0.02$, $p = .297$), and Testability ($M = +0.37$ vs. $+0.11$, $p = .373$) leaned the same way without reaching significance. Evidence Support ran counter to this pattern: improvement was greater under the baseline than under *PerspectEvolver* ($M = +0.30$ vs. $+0.79$, $p = .195$), and the within-condition improvement was significant for the baseline alone ($p = .016$).

Taken together, *PerspectEvolver* was associated with greater improvements in hypothesis novelty. Plausibility, significance, and testability moved in the same direction, although none differed significantly from the baseline. Evidence support moved in the opposite direction. This pattern may reflect a tension between novelty and existing evidentiary support [34]: a hypothesis that prior work does not already entail may also be less strongly supported by that prior work. These results suggest that supporting broader exploration may increase novelty while also increasing the need to establish an evidentiary basis for newly developed directions.

6.1.2 *Higher Usability and Lower Cognitive Load.* *PerspectEvolver* was rated higher on usability than the baseline (Figure 7; SUS: $p < .01^{**}$; per-item results in Table 4 in the appendix). Participants described the session as straightforward and quick to pick up, and said that the discussion and updated perspectives from each deliberation helped them grasp a complex multi-agent exchange at a glance. As P_8 explained, “Every step was intuitive, and I picked it up quickly.” P_{11} further noted, “There is a lot to think about when you are talking with several agents, but it was organized intuitively, and having the whole thing in view on the canvas helped a great deal.”

PerspectEvolver also significantly reduced cognitive load (Figure 7; mental demand: $p < .01^{**}$; effort: $p < .01^{**}$; frustration: $p = .109$). Participants noted that the perspectives updated without their intervention, leaving them with less to do between rounds. As P_{13} described, “Having the perspectives updated automatically was genuinely useful. It cut down the burden a lot.” P_7 added that the agents seemed to carry the earlier context with them, which made the exchange easier to stay with, “It felt like talking together and keeping pace, rather than pulling something along on my own.”

6.2 *PerspectEvolver* Achieves Design Goals: Configuring, Deliberating, and Evolving (RQ2)

In this section, we analyze system logs and think-aloud data to identify the designed affordances researchers drew on and how they read the system and its agents.

6.2.1 *Fewer Perspectives, Continually Evolved (DG1, DG3).* Under *PerspectEvolver*, participants used a mean of 3.30 perspectives (SD 1.06) from 8.40 generated candidates drawn from a mean of 1,370 retrieved papers (SD 298). Building a perspective from a cluster made the perspective’s stance easier to anticipate. As P_8 explained, “Seeing the whole thing on the cluster map and going by the representative paper for each cluster, the perspectives were easier to understand than in the baseline.” P_9 further noted, “Each paper came in six elements, so I could tell right away what the paper was saying.” In contrast, the baseline produced more perspectives ($M = 3.70$, SD 1.16). In the baseline, participants configured Perspectives directly from individual retrieved papers. This made a Perspective quick to construct when a suitable paper was available, but the resulting Perspective could be narrower or less representative. As P_{10} described, “When it

was hard to find a paper that could represent a perspective, creating that perspective was hard and I wasn't confident about it. In the deliberation I didn't really pay attention to that one."

PerspectEvolver started from fewer perspectives than the baseline, but those perspectives went through a mean of 2.95 updates (SD 2.03). Each time a thread closes, the perspectives are updated against the working synthesis and the researcher reviews the update. Over a session, then, a researcher obtained positions suited to how the discussion had developed without configuring perspectives again. As P_{12} put it, "Every time the perspectives evolved, things moved along with where the discussion was, so it felt like collaborating with a colleague. Like they already knew without me saying it." P_{14} added, "Because the perspectives evolved, they picked up the context right away, so the load was definitely lower, and I could use the AI more actively."

Overall, participants used both construction and evolution: starting from fewer perspectives, a researcher could carry a discussion through positions that fit the developing context, at a lower cost than rebuilding the panel.

6.2.2 Separate Threads, Visible Contributions, Recorded Revisions (DG2). Participants opened 42 threads in total; 27 were activated and 26 resolved, with a mean of 3.08 researcher turns per thread (SD 1.61, range = 1–7 turns). Threads ran mostly in sequence: a thread was opened before the previous one closed on 5 occasions, and 4 of 10 participants ran threads in parallel at least once. When a participant wanted one perspective's opinion, they named that perspective directly in the prompt. Interaction inside a thread took three broad shapes. Among the 26 resolved threads, 1) in 14 threads a participant drove the thread by asking perspectives directly: P_{10} put six consecutive questions to one perspective, and P_8 named three perspectives together across seven questions ($P_8, P_9, P_{10}, P_{12}, P_{13}, P_{14}$). 2) In 7 threads a participant let the panel run on its own for several turns and then began questioning on the basis of that exchange: P_{11} took two continuations before stepping in (P_9, P_{10}, P_{11}); 3) In the last 5 threads a participant lifted a specific passage from Working Synthesis and asked the authoring perspective to expand it: P_2 sent a claim about closed-loop revision failing under misleading verification feedback back to its authoring perspective for elaboration (P_2, P_3, P_7). 10 of the 26 threads addressed a single perspective, and 16 addressed more than one.

After close interaction with the perspectives, each thread closed with a working synthesis, making visible what each perspective contributed and driving revision of the hypothesis. The system proposed 91 revisions; participants accepted 64 (70.3%) as is, edited 25 (27.5%) before accepting, and rejected 2 (2.2%). Eighty-nine revisions reached the document ($M = 8.90$, $SD 4.15$, range 4–17). Methodology and Expected Results received 26 revisions each, Previous Work 21, and Framing 16, so the way a problem was approached and what it was expected to show changed more often than the problem statement itself. P_{11} mentioned "It very helpful that perspective-guided deliberation made visible what each agent had done and how it had contributed, and participants judged what to carry into the document with the source of each contribution in view." As P_2 described the loop, "With the baseline it felt like I was passively steering something, but here it felt like everyone was talking together with the same rhythm, and with more back and forth it came more naturally to stretch my thinking."

Deliberation and revision therefore worked as one loop rather than as separate stages. A participant directed the discussion by addressing specific Perspectives, the discussion closed on a working synthesis showing which perspective contributed what, and the decision to carry that into the document followed from the same view. Moving a discussion into the document required no separate step, so a participant could run the loop several times within one session.

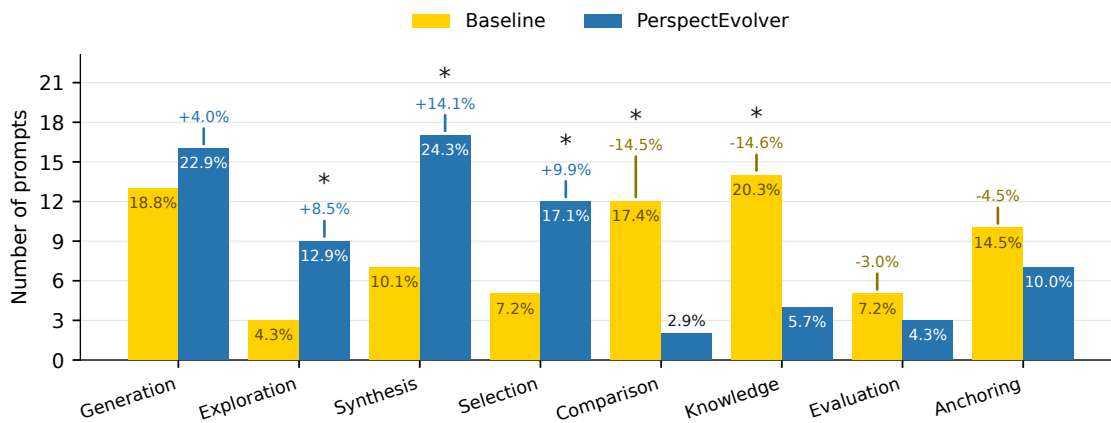


Fig. 8. Comparison of users' prompts coded with the eight creative subprocesses across conditions (* $p < 0.05$). Percentages inside each bar give that code's share of the condition's total; percentages above give the difference between conditions (positive = *PerspectEvolver* higher, negative = the baseline higher)

6.3 *PerspectEvolver* Facilitates Advancing Divergent and Convergent Thinking Activities (RQ3)

PerspectEvolver was rated as supporting both divergence and convergence (Figure 7; 3-item mean: $p < .01^{**}$; per-item results in Table 4 in the appendix). Participants explored new regions of the hypothesis space through *PerspectEvolver* and developed what they found into a hypothesis; across that process, exploration and convergence felt like one continuous movement rather than two separate efforts. P_9 described expanding and then narrowing ideas as effortless and reported that the support was most helpful during convergence, adding that the evolving perspectives were something they could feel over the session. P_{13} reported that the system helped with both divergence and convergence, and that having a direction offered was what made the difference; they also noted that seeing perspectives develop let them proceed iteratively with more context.

6.3.1 Advancing under *PerspectEvolver*, Grounding under the Baseline. To understand how participants engage in divergent and convergent thinking during system use, we analyzed prompts used in *PerspectEvolver* and the baseline (§5.3.2). As shown in Figure 8, the distribution of subprocesses differed between the two conditions. *PerspectEvolver* significantly increased EXPLORATION (4.3% vs. 12.9%, $p = .012^*$), SYNTHESIS (10.1% vs. 24.3%, $p = .004^*$), and SELECTION (7.2% vs. 17.1%, $p = .027^*$), and significantly decreased COMPARISON (17.4% vs. 2.9%, $p = .002^*$) and KNOWLEDGE (20.3% vs. 5.7%, $p = .002^*$). GENERATION rose without reaching significance (18.8% vs. 22.9%, $p = .359$), and neither EVALUATION (7.2% vs. 4.3%, $p = .875$) nor ANCHORING (14.5% vs. 10.0%, $p = .492$) differed between conditions.

Under *PerspectEvolver*, participants more often issued prompts that moved the discussion forward. For example, P_{10} asked the system SYNTHESIS, “What is the overall conclusion when you put all of your findings together?”, carrying the accounts into one statement rather than dropping any of them. After examining the alternatives and where they met, P_{10} moved to SELECTION: “Of the three angles you raised—telemonitoring intensity, high-risk subgroups, and comparison against standard follow-up—I want to focus on the last. Tell me what that comparison involves.” These moves were available to participants because they did not have to assemble the grounds for advancing by hand. Updating the perspectives and the hypothesis against the current context reduced workload (Figure 7; $p < .01^{**}$), which in turn supported planning

for the subsequent step. P_{13} reported that, “*after exploring several hypotheses, reading what the Working Synthesis had settled made the choice of direction straightforward.*” P_{12} similarly mentioned, “*The suggested threads told me exactly what the next step had to be, so moving on just felt natural.*”

By contrast, baseline participants more often issued prompts that grounded the existing direction in prior material. P_{13} first set the current direction against work already in hand (COMPARISON): “*Drift-diffusion work treats the boundary as set before the trial starts. Set my ‘revised during the task’ claim against that framing.*” Next came what the agents had not raised, supplied from personal experience (KNOWLEDGE): “*In the pilot runs I did, participants only slowed down after they had already made one visible error. Take that into account when you assess the expected results.*”

The baseline provided discussion summaries, but did not maintain a structured Working Synthesis linked to Perspective contributions or use discussion outcomes to evolve the Perspectives. Continuing the discussion therefore required participants to supply more of that grounding themselves. COMPARISON and KNOWLEDGE together accounted for 37.7% of baseline prompts against 8.6% under *PerspectEvolver*. The two conditions differed not in the number of agents, but in how the grounding needed to move the deliberation forward was represented and carried across the interaction.

7 Discussion

In this section, we reflect on what designing and evaluating *PerspectEvolver* revealed, and draw three design implications for multi-agent systems that support hypothesis generation. We then discuss the limitations of the present work and directions for future research.

7.1 Beyond Configuration: Evolving Perspectives in Human–AI Sensemaking

Sensemaking in human–AI research ideation involves both *constructing* multiple perspectives on a problem and *evolving* those perspectives as new evidence and interpretations reshape the research direction. As the research direction develops, researchers must assess how each perspective interprets the problem and whether it remains relevant to the research being pursued.

Prior multi-agent ideation systems primarily support configuring perspectives, while subsequent changes are manual or implicit [29, 30, 40]. *PerspectEvolver* makes perspective evolution inspectable by externalizing how each Perspective contributes to the developing research direction and how its interpretation shifts over time. Participants using *PerspectEvolver* achieved greater improvements in hypothesis novelty, rated the system higher in usability, and reported lower cognitive load. Unresolved disagreements and questions can redirect research by identifying aspects of the problem that require further examination. *PerspectEvolver* preserves these issues as follow-up questions, allowing outcomes from earlier deliberation to shape what researchers investigate next.

Beyond scientific research, this approach may apply to domains in which work develops through deliberation among multiple perspectives. Such settings require that (1) perspectives can be represented through interpretable elements, (2) deliberation outcomes can be traced to the perspectives that shaped them, and (3) those outcomes can revise the perspectives used as the work develops. For example, in legal counsel, (1) perspectives can be represented through issue, governing doctrine, scope, and anticipated counterargument [13]; (2) deliberation can establish a provisional position on the governing doctrine while preserving which perspectives and authorities shaped that position and leaving issues such as damage calculation unresolved; and (3) subsequent argument or evidence can revise those perspectives as unresolved issues are examined further.

7.2 Design Implication 1: Externalize Grounding, Preserve Researcher Judgment

Under *PerspectEvolver*, a larger share of participants’ coded activities involved exploring, synthesizing, and selecting ideas, while a smaller share involved comparison, prior knowledge, evaluation, and anchoring. This shift accompanied greater improvements in hypothesis novelty, suggesting that reducing the effort required to reconstruct and ground prior reasoning can leave more attention for developing the research direction.

The formative study shows what this grounding required (§3.3.3). Participants had to reconstruct what prior discussions had established, trace ideas to the perspectives that introduced them, and determine how discussion outcomes should affect the hypothesis and the perspectives. These are forms of information the system can externalize through persistent threads, contribution provenance, and linked revisions, reducing the need for researchers to recover them from conversation history [15, 19].

Ideation systems have largely focused on generating and narrowing candidate ideas. This addresses only part of the grounding required when the perspectives used to interpret those ideas also develop. When a research direction changes while those perspectives remain fixed, researchers must reconcile the developing direction with interpretations that no longer reflect it [25]. Maintaining this correspondence becomes recurring grounding work as ideation continues.

Grounding of this kind can be absorbed because such grounding is a matter of record rather than judgment: what a thread settled, where a claim originated, and which insights bear on which perspective are facts the system already holds and can keep in view. What the system should not absorb is the decision that follows—which candidate to retain, merge, or set aside—since no body of prior work determines what is worth pursuing in a researcher’s own inquiry [18]. A system that supports divergence and convergence should therefore be designed to carry the grounding that recovers what has already been produced, while leaving the researcher to advance the hypothesis.

The principle extends from the individual perspective to the ensemble. When a new thread takes up a different aspect of the research problem, the ensemble assembled for earlier threads becomes less relevant to what is now under discussion, and that mismatch is itself an occasion for grounding: the same literature that grounds a perspective’s revision can be searched for the perspectives the new thread calls for. A system that supports this loop—retrieving relevant perspectives, recomposing the ensemble for the thread that follows—would allow researchers to sustain divergent and convergent thinking with the perspectives each stage of the inquiry requires.

7.3 Design Implication 2: Friction Needs an Outlet

Prior work holds that friction in ideation systems is productive: exposing researchers to opposing views leads them to scrutinize what an agent proposes rather than accept it outright [7, 27, 30]. In design ideation, agents that challenge the dominant proposal reduce design fixation [27]; in group decision-making, devil’s-advocate agents voice minority positions against conformity pressure [7]. However, friction is not always beneficial. Cognitive forcing interventions improve the accuracy of judgment, but lower user satisfaction and trust, and counterargument agents loosen the discussion at the cost of higher cognitive load [26, 50]. More fundamentally, friction that never resolves closes off new ideation rather than opening it, producing idea fixation or premature convergence [28].

Our design gives unresolved friction a different meaning. Where the perspectives converge, the agreement enters the hypothesis; where they do not, the disagreement becomes an open question that configures the perspectives for the next round. Treating the unresolved as material for the next round makes it not a stalled state but a signal that the topic requires a new ensemble of perspectives. A system that introduces counterarguments to support diversity in ideation must therefore not only consider their strength and frequency, but also provide an outlet for the friction they produce.

Without an outlet, even a well-placed counterargument remains a burden the researcher must carry; with one, the same counterargument becomes material for the next step.

However, opening the next round automatically from what remains unresolved carries its own cost. If a researcher does not follow why the loop reopened where it did, the loop that never closes becomes a trap of a different kind from fixation [36, 48], and our system offers no account of when the space has been explored enough. Two forms of support are therefore required. The first is an agent-generated account of the provenance of each perspective revision and each suggested thread. The second is a macro-level representation of the investigation’s trajectory, through which researchers can assess whether the space has been explored sufficiently.

7.4 Design Implication 3: Extending Ensembles by Evolution Rather Than Addition

Widening the hypothesis space an ensemble of perspectives covers encourages new ideas. If perspectives can only be revised manually as an idea develops, the only way to extend the ensemble to fit the context is to add agents, and cognitive load rises accordingly [38]. Our work instead kept the number of perspectives small, supported sensemaking of how perspectives evolve, and produced hypotheses rated more novel than in the baseline.

An evolved Perspective accumulates changes across rounds while remaining a single inspectable unit. Because that unit is made explicit, each element showing what the perspective now holds, the researcher works with the unit rather than with the individual changes inside it, a form of chunking that suggests load need not scale with what each perspective has accumulated [9]. This points to a further dimension along which an ensemble can be extended: not only the number of perspectives present, but the capacity of those perspectives to develop through deliberation.

Room for improvement remains in how far sensemaking of perspective evolution reduces load. As deliberation proceeds and evolved perspectives converge, contrast weakens, and admitting a new perspective into the ensemble can raise the load again. As P_{10} mentioned, *When I bring a new agent into the discussion, it is still hard to decide what belongs in the ensemble. A recommendation that reflects the context of my deliberation would free more energy for creative decisions.* Supporting sensemaking of perspective evolution may therefore defer the point at which load arises rather than remove the load itself: while perspective evolution sustains diversity, the ensemble needs no recomposition, and the cost of selection returns only once evolution alone no longer produces contrast. For cognitive load, this points to recommending new perspectives from the context of the deliberation and the researcher’s intent, rather than leaving that selection to the researcher alone.

7.5 Limitations and Future Work

First, the duration of the user study (30 minutes) and the size of the participant pool ($N = 10$) may limit the generalizability of our findings. Participants discovered several high-quality hypotheses within the session, but in research practice a promising hypothesis takes shape over weeks and several meetings. The ten participants may also not fully represent the space of research ideation domains. For future work, we plan to collaborate with more researchers and to conduct a long-term study of at least one week to enhance the generalizability of *PerspectEvolver*.

Second, using an LLM as a judge to evaluate submitted hypotheses may introduce bias. Even though the judge’s ratings agreed with expert ratings (Krippendorff’s $\alpha = 0.75$), that agreement might not guarantee the ratings reflect what researchers in the field consider promising. Future work should complement automated evaluation with larger-scale expert assessment.

8 Conclusion

Multi-agent systems for ideation have focused on assigning diverse perspectives to agents, yet have largely left how a researcher’s perspectives evolve through deliberation to the user. We presented *PerspectEvolver*, an interactive MAS that supports the sensemaking required to configure and evolve a perspective ensemble: each perspective is built from a cluster of literature and represented as six inspectable elements, deliberation runs in topic threads that condense into a statement of what settled and what remains open, and that statement revises the hypothesis and the perspectives while what no thread settled opens the round that follows. In a within-subjects study with 10 science and engineering researchers, *PerspectEvolver* produced greater improvements in hypothesis novelty and was rated higher in usability and support for divergent and convergent thinking, while participants reported lower cognitive load. Advancing utterances also accounted for a larger share of the session under *PerspectEvolver*, suggesting that effort otherwise spent recovering and organizing what a discussion had produced was available instead for moving the inquiry forward. We hope this work encourages the design of multi-agent systems in which perspectives evolve explicitly as deliberation unfolds.

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Fig. 9. Baseline interface used in the user study. (a) *Input*. Participants begin from a research problem and a four-part initial hypothesis. (b) *Perspective configuration with papers*. Participants configure paper-grounded Perspectives by selecting retrieved papers and editing each Perspective’s job and description before adding it to the panel. (c) *Group-chat based deliberation with multi-agent*. Participants deliberate across three suggested topics with support for direct messages, configurable agent turns, adding or removing Perspectives, and constructing additional Perspectives during the discussion. Discussion summaries can be transferred into the four Notepad sections, whose versions are maintained independently.

A Appendix

A.1 Codebook Development

To code participants’ prompts, we drew on two studies of creative subprocesses:

- **Boldt [2] (8)**: generation, selection, evaluation, exploration, synthesis, goal refinement, comparison to other creative works, application of domain-specific knowledge
- **Boldt and Kaufman [3] (8)**: generation, elaboration, association, selection, evaluation, anchoring, metacognition, filler

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- **Union** (13): generation, selection, evaluation, exploration, synthesis, goal refinement, comparison, domain knowledge, elaboration, association, anchoring, metacognition, filler¹⁴

We adapted the code definitions to prompts written to an AI system during scientific hypothesis generation.

A.2 Interview Questions

- Can you walk me through your overall experience developing your hypothesis in each session?
- How did you find the process of building perspectives and bringing them in and out of the discussion?
- Did you notice the perspectives changing over the session? If so, what did you make of that?
- Were there points where the session changed direction? What happened around those?
- What helped, and what got in the way, as you developed your research idea?

A.3 Five dimensions for evaluating hypotheses

We evaluated the quality of submitted hypotheses on the five dimensions with the LLM judge (GPT-5.6 Luna) below,

- **Novelty**: the degree to which the hypothesis is not already entailed by prior work Si et al. [45].
- **Plausibility**: whether the hypothesis is consistent with established scientific principles and theories [12].
- **Significance**: the potential importance of the hypothesis if validated Yang et al. [49].
- **Testability**: whether the hypothesis can be practically tested with available techniques, equipment, and resources [12].
- **Evidence Support**: the extent to which the hypothesis is grounded in, or verifiable against, existing evidence Ke et al. [21].

A.4 Technical Details of PerspectEvolver

PerspectEvolver comprises Perspective Agents, the Deliberation Space, and the Deliberation Protocol.

A.4.1 Perspective Agents. A **Perspective Agent** is a research collaborator defined by persistent Perspective state and shared generation modules.

- **Role.** Develops ideas, addresses research questions, and distinguishes evidence from interpretation.
- **Perspective.** A scientific focus, *Framing*, *Position*, and PerspectiveFragments for *Scope*, *Explanation*, *Approach*, and *Significance*.
- **Knowledge.** Source-linked papers, passages, and Literature Evidence.
- **Working Context.** The current Perspective, Investigation Brief, thread context, and researcher request.
- **Interaction History.** Prior Contributions, retrieved evidence, syntheses, and revisions.
- **Interaction Protocol.** Rules for participation and response within the Perspective Ensemble.
- **Model and Tools.** Language-model, retrieval, and reranking components.

Implementation. Agents share DSPy modules and load the latest Perspective before each action. openai/gpt-5.6-luna performs search planning, evidence extraction, Perspective naming, proposal generation, deliberation, direct replies, evidence requests, annotations, and local rewriting. openai/gpt-5.6-terra performs Perspective synthesis, Working Synthesis, Reflection, document revision, and follow-up generation.

¹⁴The last two codes from Boldt [2] are shortened here and in the remainder of the paper.

Table 4. Post-session items rated for both conditions on a 7-point Likert scale: three NASA-TLX demand items, ten System Usability Scale items, and three items on divergent and convergent support.

Item	Baseline		PerspectEvolver		<i>p</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
<i>NASA-TLX</i>					
How mentally demanding was the task? (R)	5.70	0.67	3.60	1.71	.004**
How hard did you have to work to accomplish your level of performance? (R)	4.30	1.70	2.00	0.94	.008**
How insecure, discouraged, irritated, stressed, and annoyed were you? (R)	4.10	1.52	2.90	1.45	.109
<i>System Usability Scale</i>					
I think that I would like to use this system frequently	3.40	1.78	6.00	0.67	.008**
I found the system unnecessarily complex (R)	3.80	1.75	3.30	1.95	.547
I thought the system was easy to use	4.20	1.87	6.20	0.63	.008**
I think that I would need the support of a technical person to be able to use this system (R)	3.90	1.52	3.80	2.10	.945
I found the various functions in this system were well integrated	3.60	1.35	6.10	0.57	.004**
I thought there was too much inconsistency in this system (R)	2.80	1.23	1.90	1.20	.109
I would imagine that most people would learn to use this system very quickly	4.70	1.25	5.90	0.99	.055
I found the system very cumbersome to use (R)	4.30	1.89	1.80	1.03	.008**
I felt very confident using the system	3.40	1.90	5.80	0.79	.016*
I needed to learn a lot of things before I could get going with this system (R)	4.20	1.62	2.60	1.71	.055
<i>Divergent / convergent support</i>					
To what extent does this system extend the boundary of hypothesis generation by facilitating the exploration of multiple perspectives?	2.60	0.97	6.20	0.63	.002**
To what extent does this system help narrow toward a promising direction by assisting in evaluating perspectives?	2.70	0.95	5.80	0.79	.004**
To what extent does this system support iterative cycles of exploring a range of candidate hypotheses and narrowing to the promising ones?	2.70	1.25	5.70	0.82	.002**

Discovery and Formation. Semantic Scholar snippet search retrieves papers from Investigation Brief queries [23]. Normalized SPECTER2 embeddings [46] are reduced with UMAP [33] and clustered with HDBSCAN [32], with a separate two-dimensional UMAP projection for the Literature Map. Representative papers are reranked with voyageai/rerank-2.5. Evidence extraction retains only Literature Evidence whose supporting quotation occurs verbatim in its mapped source passage. Perspective synthesis generates *Framing*, *Position*, and Perspective Fragments, and validation requires every nonempty field to reference valid Literature Evidence. Before deliberation, voyageai/rerank-2.5-lite reranks Literature Evidence from the Perspective and discussion for the current thread and researcher request.

A.4.2 Deliberation Space.

- **Investigation.** The research problem, literature corpus, Investigation Brief, selected Perspectives, and research artifacts.
- **Perspective Ensemble.** The researcher-selected Perspective Agents available to the Investigation.
- **Deliberative Thread.** A persistent discussion with its research question, Perspectives, Contributions, evidence requests, synthesis history, and origin references.

Implementation. The Deliberation coordinator stores discussion state by thread while Perspective state persists across threads. Thread references preserve branches from Proposals and prior discussions, while Contribution targets preserve

replies. Researcher actions resume checkpointed LangGraph workflows, and follow-up questions create threads linked to their originating discussion and Working Synthesis.

A.4.3 Deliberation Protocol. The **Deliberation Protocol** governs turns, response targets, evidence requests, and synthesis.

- **DeliberationPolicy.** Speaker and target selection and stopping conditions.
- **DiscussionRound.** A researcher-requested sequence of agent turns.
- **TurnAction.** One action with its Perspective, target, instruction, and optional MoveType.
- **Contribution.** An authored response with scientific statements, target, type, quotation, and evidence references.
- **EvidenceRequest.** An unresolved empirical question and literature query.

Implementation. Discussion rounds select speakers and targets until the requested turns complete or the exchange stops. An EvidenceRequest searches Perspective literature, the investigation corpus, and external literature when configured, then extracts source-linked Literature Evidence. A WorkingSynthesis uses Contributions, cited Literature Evidence, and evidence-request outcomes. The system extracts provenance-linked DiscussionInputs that distinguish empirical findings from scientific considerations. Reflection compares these inputs with each Perspective and returns no_change, update_position, or update_framing_and_position; proposed changes retain their motivating inputs and previous content. Investigation Brief revisions and follow-up questions are generated separately from the Working Synthesis, with LangGraph pausing for researcher review before document changes are applied.